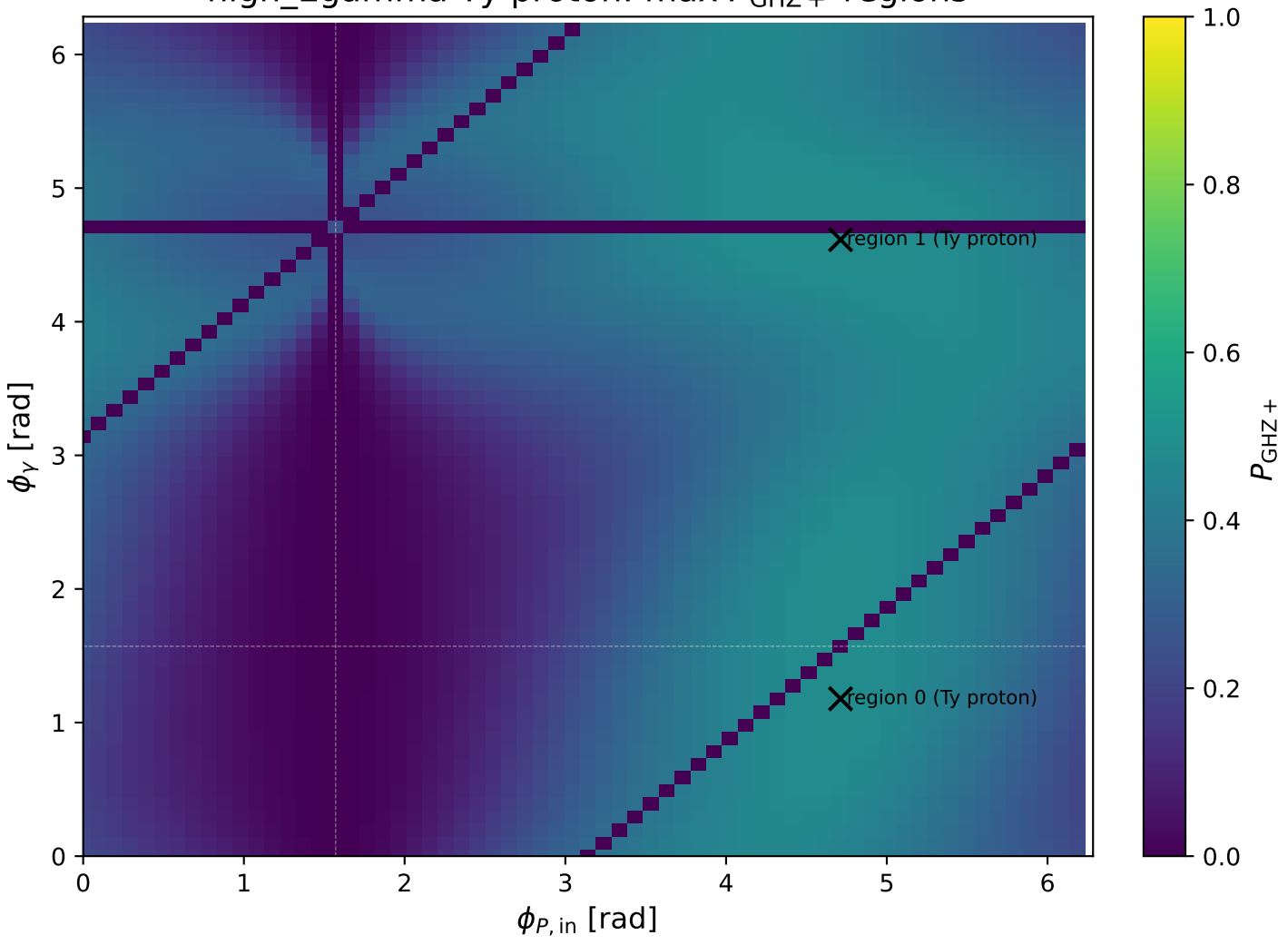
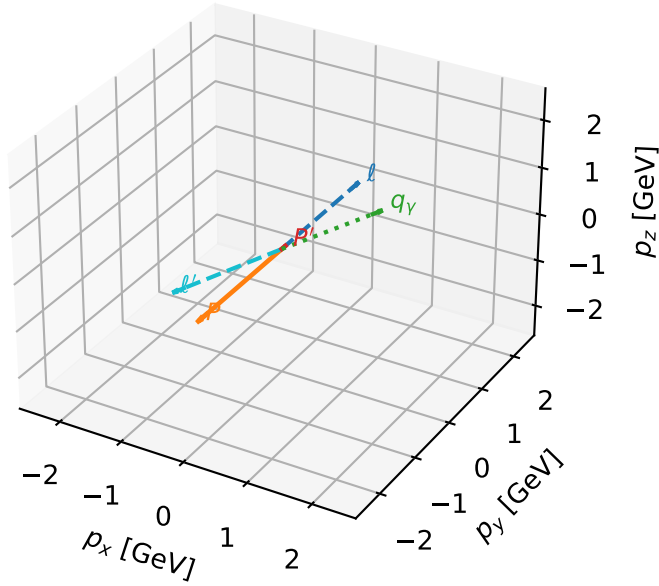


high\_Egamma Ty proton: max  $P_{\text{GHZ}+}$  regions



# Ty proton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta



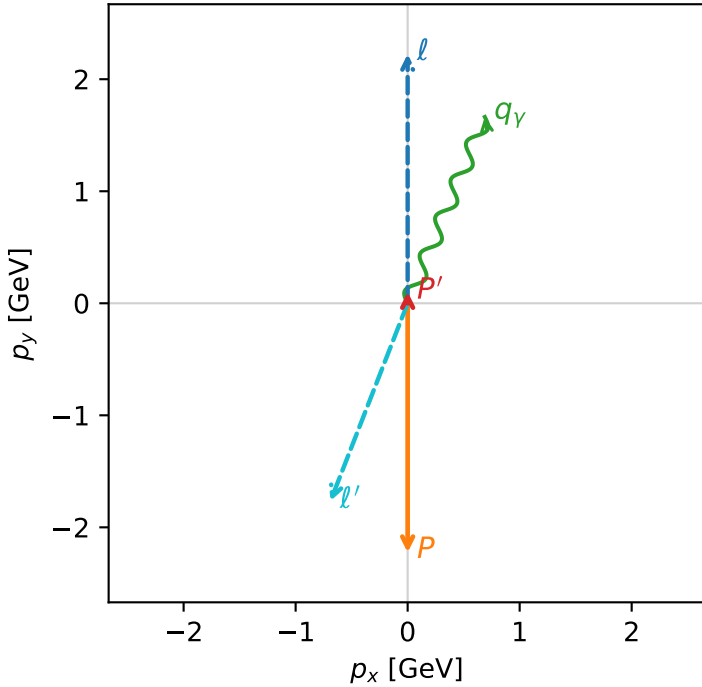
ghz\_purity\_high\_egamma\_ty\_proton\_region\_0  $P_{\{\text{rm GHZ}\}}=0.482724$   
 region: high\_Egamma\_Ty\_proton\_region\_0  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e}\rho(h_e, T_y, p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.102337$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=1.5708$ ,  $\phi_p=4.71239$   
 $E_{\gamma}=1.8$ ,  $\phi_{\gamma}=1.1781$   
 $Q^2=16.2839$ ,  $x_B=0.841965$ ,  $t=-3.25132$ ,  $W^2=3.93631$ ,  $y=0.937217$   
 $\theta(\ell', \gamma)=178.816$  deg

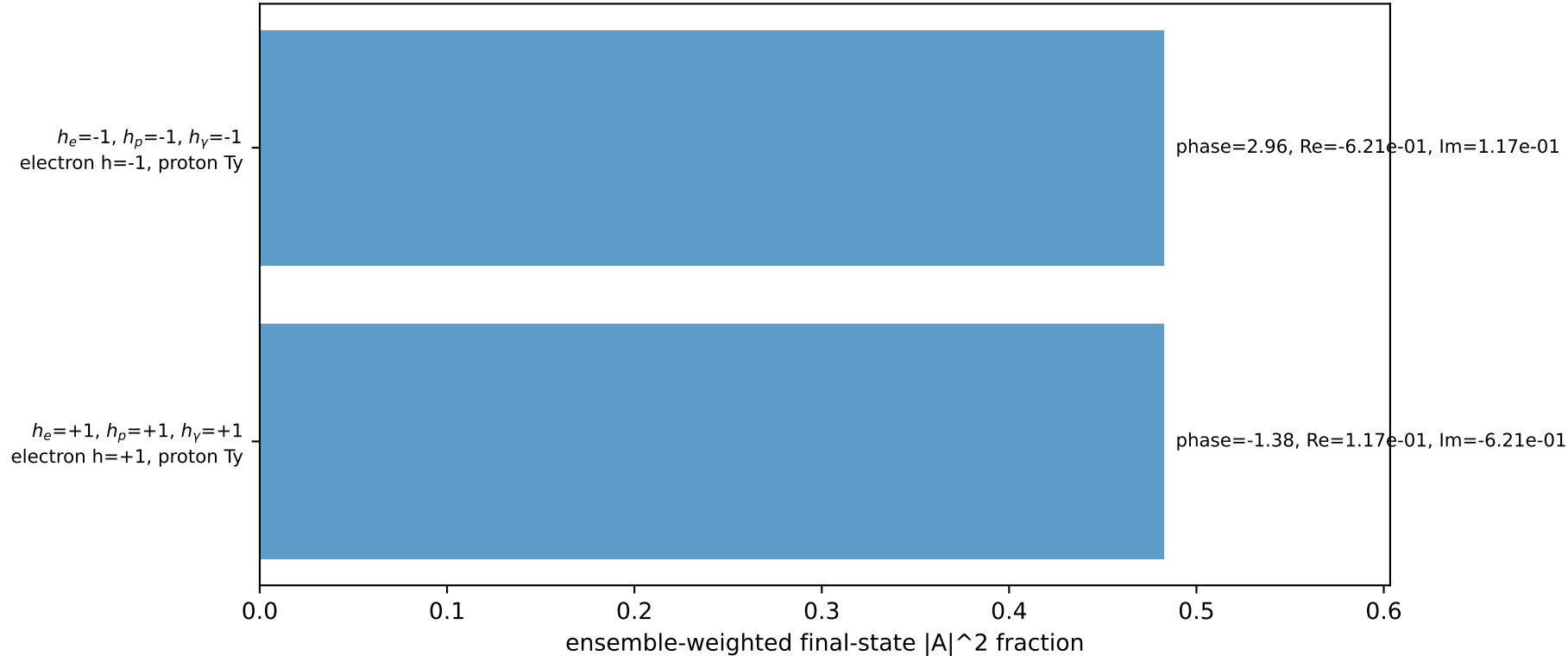
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8950$   $p_x= -0.6888$   $p_y= -1.7653$   $p_z= 0.0000$   
 $P'$   $E= 0.9436$   $p_x= 0.0000$   $p_y= 0.1023$   $p_z= 0.0000$   
 $q_{\gamma}$   $E= 1.8000$   $p_x= 0.6888$   $p_y= 1.6630$   $p_z= 0.0000$

Transverse plane

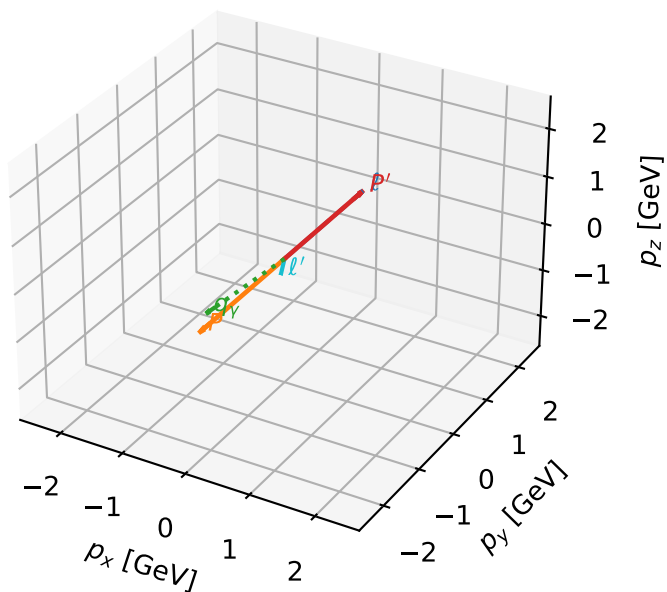


Leading initial-ensemble/final-state components (N=2, retained=96.5%)



# Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

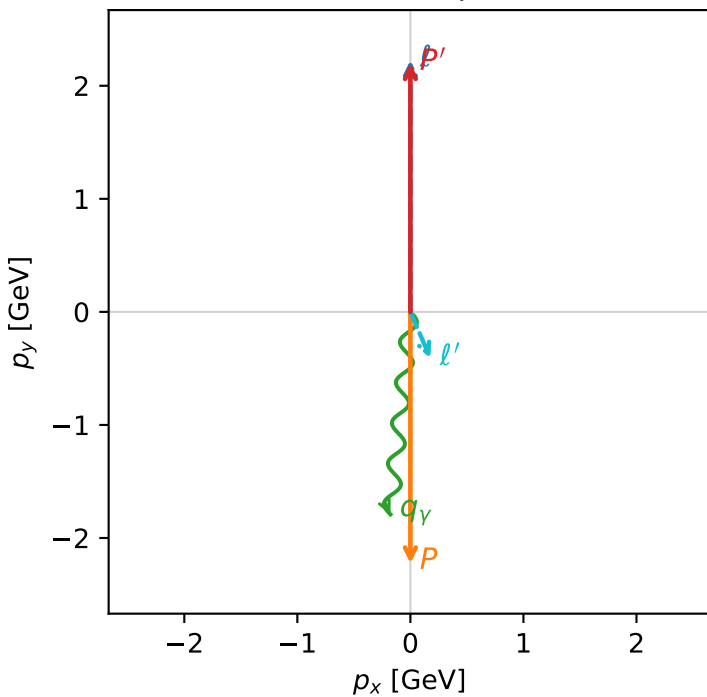


ghz\_purity\_high\_egamma\_ty\_proton\_region\_1  $P_{\text{GHZ+}}=0.482457$   
 region: high\_Egamma\_Ty\_proton\_region\_1  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e} \rho(h_e, T_{y,p})$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.20096$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_V=1.8$ ,  $\phi_V=4.61421$   
 $Q^2=3.80661$ ,  $x_B=0.187472$ ,  $t=-19.5835$ ,  $W^2=17.3782$ ,  $y=0.983958$   
 $\theta(l', \gamma)=28.9269$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $l'$   $E= 0.4460$   $p_x= 0.1764$   $p_y= -0.4096$   $p_z= 0.0000$   
 $P'$   $E= 2.3925$   $p_x= 0.0000$   $p_y= 2.2010$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -0.1764$   $p_y= -1.7913$   $p_z= 0.0000$

Transverse plane



$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=-1$   
 electron  $h=-1$ , proton  $\bar{\gamma}$

$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=+1$   
 electron  $h=+1$ , proton  $\bar{\gamma}$

Leading initial-ensemble/final-state components (N=2, retained=96.5%)

